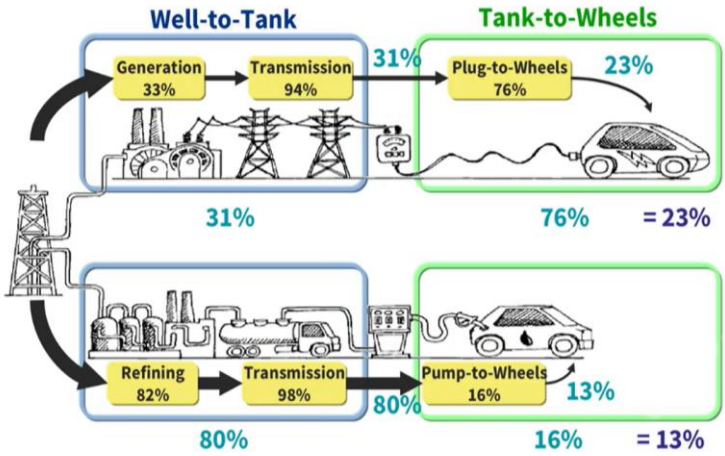
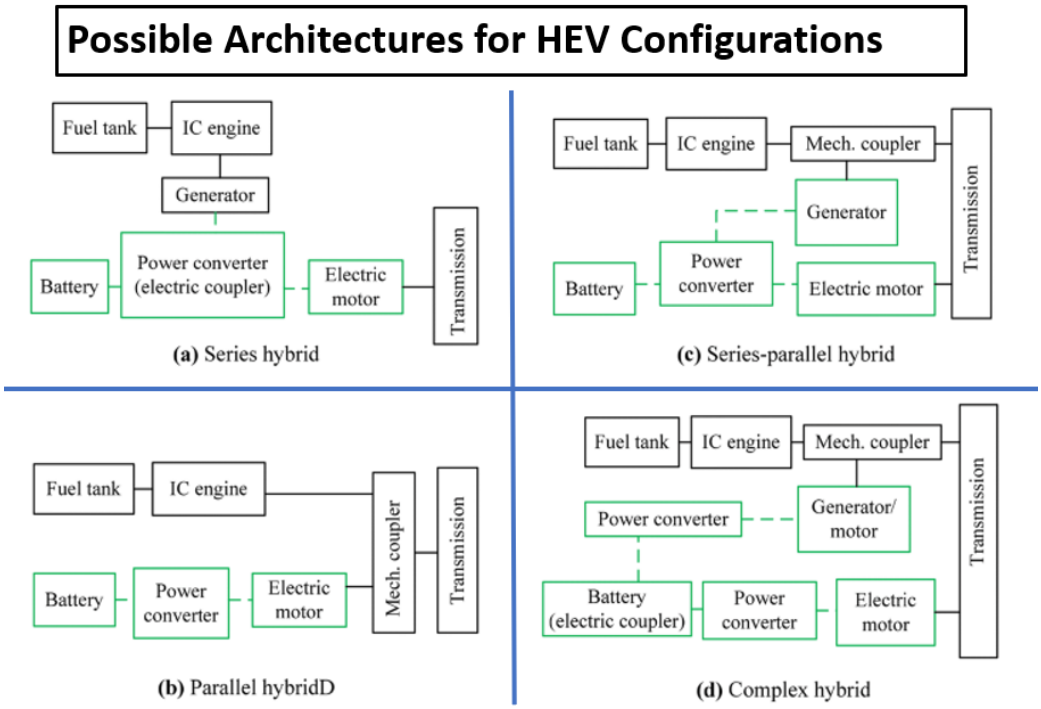
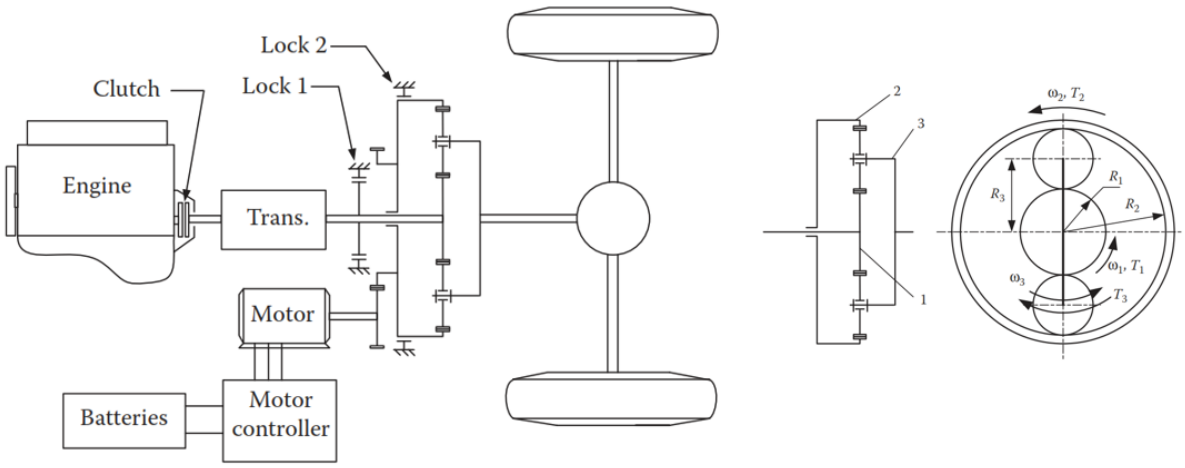
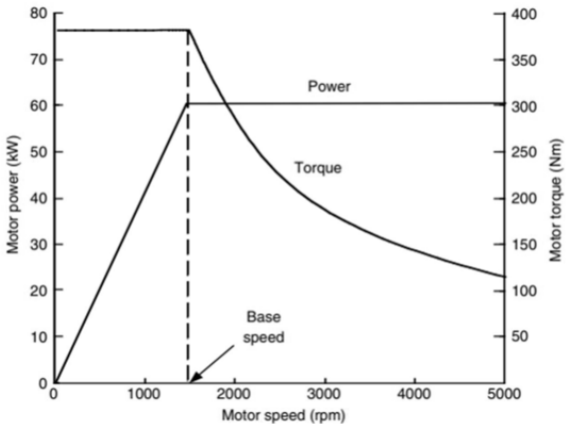
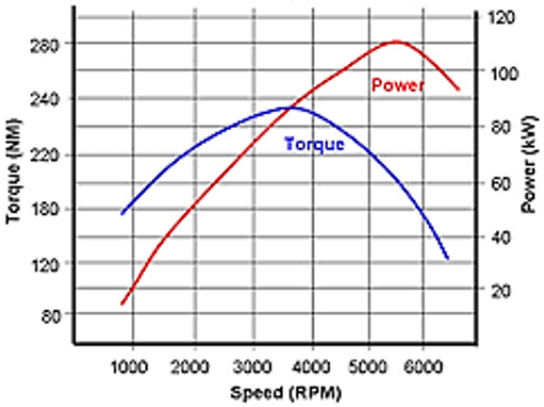
MANIPAL INSTITUTE OF TECHNOLOGY
MANIPAL
(A constituent unit of MAHE, Manipal)

DEPARTMENT OF Aeronautical and Automobile Engineering			
END SEMESTER EXAMINATION		NOV-DEC, 2024	
Level	**B.Tech**	**Curriculum**	**2022**
Program Name	**Automobile Engineering**	**Course Type**	**PE**
Course Name	**Hybrid and electric Vehicles**	**Course Code**	**AAE 4039**
Semester	**VII**	**Max Marks**	**50**
Date	**29-11-2024**	**Duration**	**3 hrs**

Q. No.	Question	Marks
1A	A 4-wheeler accelerates from 0 to 60 kmph in 30 s. It then travels at a constant speed of 60 kmph for 10 min. it then decelerates to 0 kmph in 30 s. Compute the energy used in Wh/km, assuming R=1 and R= 0.5 and the distance traveled. Details of vehicle: GVW (kg) =1600 Cd =0.3 Area (Sq m)= 2.5 μ = 0.015 Tyre radius = 0.3 m **Solution:**	4
1B	Illustrate and compare the wheel-to-well efficiencies of electric vehicles (EVs) and internal combustion (IC) engine vehicles using labeled diagrams.	3

	 Illustration 2 marks Explanation 1 mark	
1C	Analyze the economic and environmental impacts of electric vehicles (EVs) and evaluate their role in promoting sustainable development. Economic impact: **Automotive industry transformation** **1 mark** Manufacturing shift: Traditional automakers are investing heavily in EV production, leading to a reconfiguration of manufacturing processes and supply chains. New entrants: Companies such as Tesla, Rivian and Lucid have emerged, creating competition and innovation. Job displacement and creation: Jobs in traditional auto manufacturing may decline, but new opportunities are emerging in battery production, software development, and EV assembly. **Energy sector dynamics** **1 mark** Reduction in oil demand: EV adoption reduces demand for gasoline and diesel, potentially lowering global oil prices and impacting oil-exporting economies. Increased electricity demand: Utilities see increased demand for electricity, leading to investment in grid infrastructure and renewable energy sources. Investment in charging infrastructure: Billions will be invested in charging networks, creating new business opportunities and jobs. **Environmental and health benefits** **1 mark** Reduced emissions: EVs help reduce greenhouse gas emissions, improve air quality and potentially reduce pollution-related healthcare costs. Economic value of sustainability: Companies and governments benefit economically by meeting climate targets and avoiding penalties for exceeding emissions limits.	**3**

2A	Illustrate and explain the various hybrid electric vehicle (HEV) architectures using clear and labeled diagram  Illustration 2 marks Explanation: 2 marks	**4**
2B	With a neat sketch, discuss the working of a Hybrid electric drivetrain with speed coupling using a planetary gear unit.  	**3**

	1. *Hybrid traction:* When lock 1 and lock 2 are released (the sun gear and the ring gear can rotate), both the engine and the electric machine supply positive speed and torque (positive power) to the drive wheels. The output speed and torque from the yoke of the planetary unit are described by Equations 6.11 and 6.13. That is, the rotational speed of the yoke is the summation of the sun gear speed (engine speed, or proportional to engine speed) and the ring gear speed (electric motor speed, or proportional to motor speed). However, the output torque from the yoke is proportional to the engine torque and the motor torque. Torque control will be studied 2. *Engine-alone traction:* When lock 2 locks the ring gear to the vehicle frame and lock 1 is released, the engine alone supplies power to the drive wheels. From 3. *Motor-alone traction:* When lock 1 locks the sun gear to the vehicle frame (engine is shut off and engine clutch is disengaged) and lock 2 is released, only the electric motor supplies its power to the drive wheels. From Equations 6.11 and 6.13, 4. *Regenerative braking:* The states of lock 1 and lock 2 are the same as in motor-alone traction, the engine is also shut off, the engine clutch is disengaged, and the electric machine is controlled in regenerating mode (negative torque). The kinetic or potential energy of the vehicle can be absorbed by the electrical system. Figure 1.5 marks Explanation 1.5 marks	
2C	Illustrate and compare the traction characteristics of a DC traction motor and an internal combustion engine (ICE) using a detailed and labeled diagram. **Traction motor Characteristics**  **Figure shows variable speed electric motor** **By definition:** **Power = Torque X angular Speed** **Compared with the IC engine** **Engine Power and Torque Curves**  **Torque speed characteristics for IC engine** Graphs 2 marks Explanation 1 mark	**3**

3A	With a neat sketch explain how a rotating magnetic field is achieved in an induction motor using a balanced 3-phase current.	**3**
3B	Using an electrical model of a motor deduce the relationship between motor rpm (ω_m), torque (τ) and armature current (i_a), and voltage (v_a), respectively.	**4**
3c	With the help of an equivalent circuit explain the working of the buck converter in on-state and off-state.	**3**
4A	Discuss the components involved and the circuit used in full-wave rectification used in the conversion of AC to DC.	**3**
4B	List the technical safety measures used in battery technology to ensure the safety of a battery pack.	**3**
4C	Discuss how dynamic braking is achieved using MOSFET with the help of a detailed 2-quadrant chopper circuit.	**4**
5A	Discuss the construction and working of fuel cell hybrid drive train design	**4**
5B	Write a short note on control strategies and design principles of series hybrid vehicles.	**3**
5C	What are ultracapacitors? Discuss the principle of operation for the same	**3**